



Documentation

Portfolio Optimization Engine

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I Introduction

This report documents the end-to-end development of a quantitative long-only equity strategy backtested over the decade spanning January 2010 to December 2020. The objective was to design a strategy that outperforms the Nifty 500 benchmark on a risk-adjusted basis while maintaining investment discipline through stock selection, weight allocation, and risk mitigation.

The development process was *iterative*: each approach exposed shortcomings that motivated the next. Beginning from a clean implementation of classical **Mean-Variance Optimization (MVO)**, we progressively added dynamic re-balancing logic, composite stock-selection via scoring, **Relative Strength (RS)** analysis, and a modified stock selection mechanism. The final strategy synthesizes all these components into a coherent, fully systematic pipeline that achieves a **CAGR of 37.51%** versus the benchmark CAGR of 8.28%, with a **Sharpe Ratio of 1.59** and a maximum drawdown of -26.96% .

The remainder of this report is structured as follows. Section 2 formally states the problem. Section 3 details each experimental approach in chronological order of development. Section 4 describes the final, unified strategy. Section 5 presents a consolidated discussion of results, and Section 6 concludes with future directions.

2 Problem Formulation

2.1 Investment Universe and Data

The investment universe given to us consisted of a set of 500 stocks from the period January 2010 to December 2020. Daily OHLCV (Open, High, Low, Close, Volume) data for each stock, along with GICS sector labels and market capitalisation labels, constitutes the primary data source. The **execution price** for each period is computed as the OHLC mean:

$$P_t^{\text{exec}} = \frac{O_t + H_t + L_t + C_t}{4}, \quad (1)$$

which provides a practical estimate of the average tradeable price, more conservative than the open or close.

2.2 Portfolio Optimization Framework

Instead of actively trading in stocks using technical indicators, as in swing trading, we formulated the problem as a portfolio optimization framework. The optimizer solves for a weight vector $\mathbf{w} \in \mathbb{R}^N$, where N is the number of stocks in the current investable universe and $\mathbf{w}$ is the proportion of total capital invested in each particular stock. For a lookback window of T trading days, the vector of annualized expected returns and the annualized covariance matrix are estimated as:

$$\hat{\boldsymbol{\mu}} = 252 \cdot \bar{\mathbf{r}}, \quad (2)$$

$$\hat{\boldsymbol{\Sigma}} = 252 \cdot \text{Cov}(\mathbf{R}), \quad (3)$$

where $\mathbf{R}$ is the $T \times N$ matrix of daily log-returns over the lookback window and $\bar{\mathbf{r}}$ is the column-mean vector. The general constrained optimization problem is then formulated as:

$$\min_{\mathbf{w}} f(\mathbf{w}) \quad (4)$$

$$\text{subject to: } \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \geq 0 \quad \forall i, \quad (5)$$

with the objective function $f(\mathbf{w})$ specialized to different criteria as discussed in the experiments section. In the final strategy, additional box constraints $w_{\min} \leq w_i \leq w_{\max}$ are imposed to prevent excessive concentration.

3 Experimental Approaches

The strategy was developed through five incremental approaches, each building upon the findings of its predecessor and addressing its shortcomings. The following subsections document each step in detail.

3.1 Approach 1: Mean-Variance Optimization with Multiple Objectives

3.1.1 Motivation and Theoretical Background

The natural starting point for any portfolio construction is **Modern Portfolio Theory (MPT)**, developed by Markowitz (1952). The central insight of MPT is that rational investors care not only about expected return but also about the variance of returns; diversification across imperfectly correlated assets reduces portfolio variance for a given expected return.

Given the estimated moments in Equation (3), three canonical optimization objectives were examined on the full universe for the period 2010–2020, with a rolling 252-day lookback:

1. Maximum Sharpe Ratio (Max Sharpe). The objective is to maximize the *ex ante* Sharpe ratio:

$$\min_{\mathbf{w}} - \frac{\hat{\boldsymbol{\mu}}^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \hat{\boldsymbol{\Sigma}} \mathbf{w}}}. \quad (6)$$

2. Maximum Return. The objective is to maximize expected portfolio return:

$$\min_{\mathbf{w}} - \hat{\boldsymbol{\mu}}^\top \mathbf{w}. \quad (7)$$

3. Minimum Volatility (Min Vol). The objective is to minimize portfolio variance:

$$\min_{\mathbf{w}} \mathbf{w}^\top \hat{\boldsymbol{\Sigma}} \mathbf{w}. \quad (8)$$

All three formulations were solved using Sequential Least Squares Programming (SLSQP) with long-only constraints ($w_i \geq 0, \sum_i w_i = 1$).

3.1.2 Results

Figure 1 presents the cumulative equity curve and drawdown plots for all three MVO objectives. The comparison between the three frameworks is presented in Table 1.

Table 1: Performance Metrics Summary for MVO Objectives (2010–2020, buy-and-hold, no rebalancing).

Objective	Ann. Return	Ann. Vol	Sharpe	Max DD	Final Rs.1 →	CAGR
Max Sharpe	17.96%	14.92%	1.20	−21.06%	Rs. 4.55	16.66%
Min Vol	15.62%	13.91%	1.12	−20.40%	Rs. 3.79	14.50%
Max Return	18.70%	15.71%	1.19	−21.63%	Rs. 4.79	17.24%

3.1.3 Discussion and Shortcomings

The **Max Sharpe** strategy achieved the best risk-adjusted profile with a Sharpe of 1.20, outperforming Min Vol (1.12) and narrowly beating Max Return (1.19), despite Max Return achieving a marginally higher return.

However, several shortcomings became apparent upon closer inspection. First, the strategies were constructed *once* at the start of the period and held without any rebalancing. Market conditions evolve substantially over a decade, and a static weight vector cannot adapt to changing volatility regimes. Second, the optimization was performed over the *entire universe* without any pre-filtering step or stock selection; this makes the data prone to noise due to high dimensionality. Third, and most critically, the approach entirely ignores stock-level quality signals such as momentum, liquidity, and trend consistency. These limitations motivated the second experimental approach.

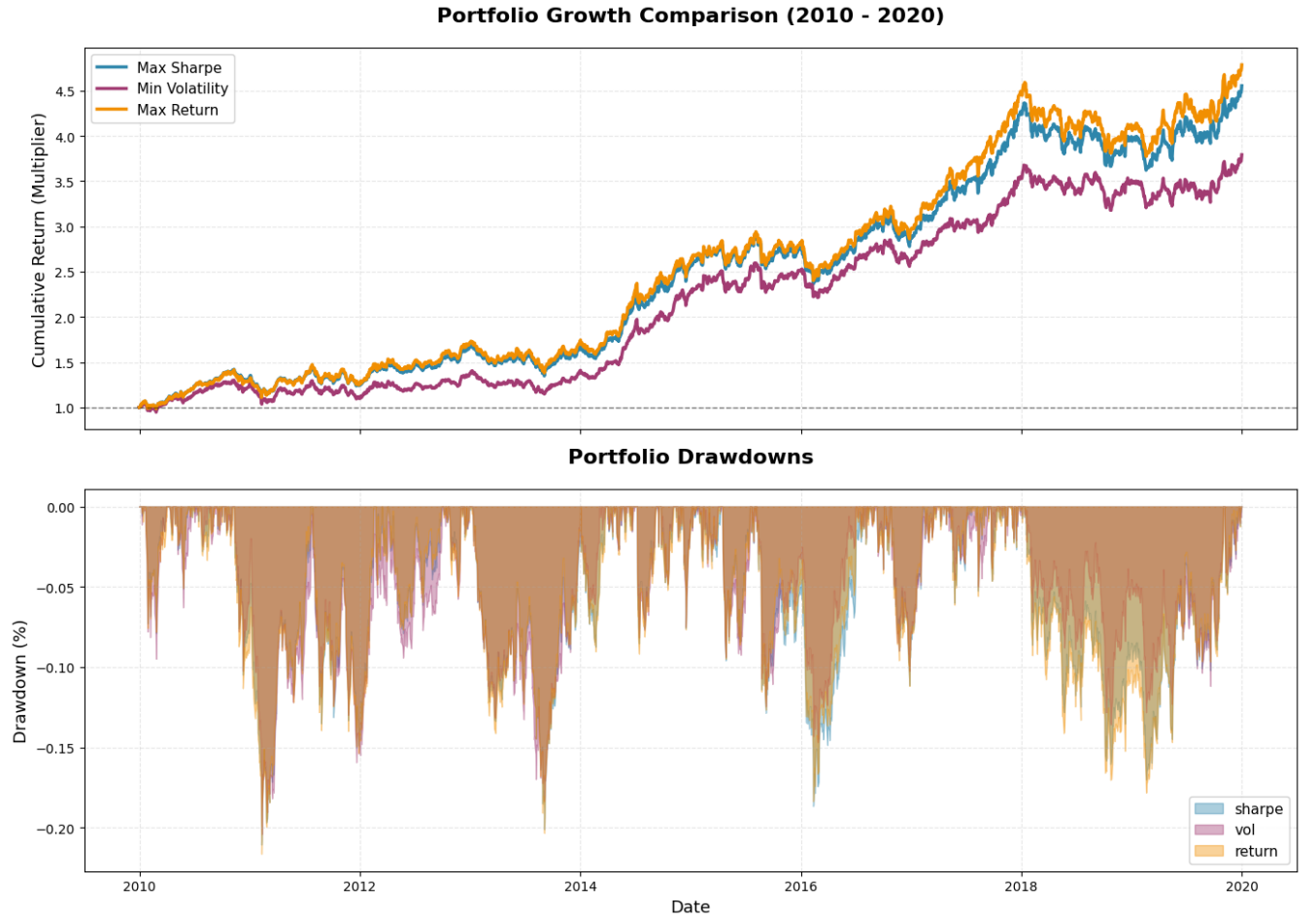


Figure 1: MVO Objective Comparison (2010–2020). *Top:* Cumulative return multiplier for the three MVO objectives starting at a base of 1.0. The Max Return strategy marginally leads in terminal wealth (Rs. 4.79), closely followed by Max Sharpe (Rs. 4.55), while Min Volatility trails at Rs. 3.79. *Bottom:* Overlaid drawdown profiles illustrating that Min Volatility, despite lower absolute returns, also achieves the shallowest worst drawdown (−20.40%).

3.2 Approach 2: MVO with Rebalancing (Static and Volume-Triggered)

A static MVO portfolio drifts away from its initial optimal composition as individual stock prices move. Periodic rebalancing restores target weights back to optimal and keeps pace with the evolving market.

3.2.1 Rebalancing Mechanisms

Two rebalancing schemes were tested under the Max Sharpe objective:

1. Static Rebalancing: Portfolio weights are recomputed and rebalanced every 30 calendar days. This is the simplest possible dynamic scheme.

2. Volume Spike + Static Rebalancing: An additional trigger is introduced: whenever the average daily traded volume of the current portfolio holdings exceeds $1.35 \times$ its 60-day historical baseline, the portfolio is flagged for immediate rebalancing. The intuition is that abnormal volume often precedes significant price movements, and early repositioning can capture the subsequent momentum or limit drawdown from a deteriorating position. Formally, the volume-spike trigger fires on day t if:

$$\bar{V}_t^{(5)} > \theta \cdot \bar{V}_t^{(60)}, \quad (9)$$

where $\bar{V}_t^{(5)}$ is the five-day rolling average portfolio volume, $\bar{V}_t^{(60)}$ is the sixty-day baseline, and $\theta = 1.35$ is the spike threshold. A minimum inter-rebalance gap of seven days is enforced to prevent excessive turnover.

3.2.2 Results

Figure 2 presents the cumulative equity curves for all three rebalancing schemes, and Table 2 summarizes the key metrics.

Table 2: Performance Metrics Comparison — Max Sharpe Strategy with Rebalancing Variants (2011–2020).

Strategy	Ann. Return	Ann. Vol	Sharpe	Max DD	CAGR
Buy-and-Hold	16.32%	14.50%	1.13	−17.37%	15.10%
Static (30-day)	19.21%	14.68%	1.31	−21.49%	17.92%
Volume Spike + Static	20.63%	14.55%	1.42	−21.15%	19.35%



Figure 2: Rebalancing Strategy Comparison — Max Sharpe Portfolio (2011–2020). The progressive improvement from passive holding to dynamic rebalancing is clearly visible, with the combined volume-spike strategy pulling decisively ahead after 2014, particularly during the high-volatility regime of 2018–2019. Vertical red lines mark volume-triggered rebalancing events, illustrating how they cluster around periods of market stress when repositioning is most valuable.

3.2.3 Discussion and Shortcomings

Static rebalancing improves the Sharpe from 1.13 to 1.31, adding 2.89% of annualized return with the same volatility. Adding volume-spike triggers further raises the Sharpe to **1.42** and the CAGR to **19.35%**, confirming that responsiveness to abnormal market activity captures alpha.

Critically, the **volume-triggered** strategy also reduces the maximum drawdown marginally (from −21.49% to −21.15%) relative to static rebalancing, suggesting that early repositioning during high-volume stress events provides some degree of tail-risk mitigation.

Nevertheless, a fundamental limitation remained: the strategy still allocated across the *entire universe* based solely on covariance-estimated portfolio weights. It does not include a pre-filtering step or stock-level quality signals. The next step was therefore to introduce an active stock-selection layer.

3.3 Approach 3: Score-Based Stock Selection (VAM, Trend, Liquidity)

3.3.1 Motivation

Finance literature identifies several robust cross-sectional return predictors. **Momentum** and **Trend consistency** (Jegadeesh and Titman, 1993) differentiate stocks with steady directional price appreciation from those exhibiting high-volatility choppiness, while **Liquidity** (Amihud, 2002) captures the risk premium associated with trading friction.

This approach constructs a *composite stock-selection score* based on volatility, momentum, and liquidity that ranks all stocks on each rebalancing date and retains only the top-ranked stocks for the MVO step.

3.3.2 Feature Construction

The following features are computed per stock on a rolling basis:

Volatility-Adjusted Momentum (VAM) Score. Captures the quality of momentum — meaning high returns achieved with relatively low volatility:

$$\text{VAM}_t = \frac{r_t^{(20)}}{\sigma_t^{(20)}}, \quad (10)$$

where $r_t^{(20)}$ is the 20-day return and $\sigma_t^{(20)}$ is the rolling 20-day standard deviation of daily returns. A high VAM indicates a stock that has risen sharply with relatively contained volatility.

Trend Score. Measures how far the current price lies above its 20-day simple moving average:

$$\text{Trend}_t = \frac{C_t - \text{SMA}_{20,t}}{\text{SMA}_{20,t}}, \quad (11)$$

where C_t is the closing price. A positive trend score indicates the stock is in a near-term uptrend.

Liquidity Score. Based on the ratio of traded value to market capitalisation, smoothed with a 20-day rolling average:

$$\text{Liq}_t = \left(\frac{\text{Traded Value}_t}{\text{MCap}_t} \right)_{(20)}, \quad (12)$$

A higher liquidity score indicates a more actively traded stock relative to its size.

Composite Score. The three signals are linearly combined with empirically tuned weights:

$$Q_t = w_{\text{VAM}} \cdot \text{VAM}_t + w_{\text{Trend}} \cdot \text{Trend}_t + w_{\text{Liq}} \cdot \text{Liq}_t, \quad (13)$$

with $w_{\text{VAM}} = 0.5$, $w_{\text{Trend}} = 5.0$, and $w_{\text{Liq}} = -2.0$. The *negative* weight on liquidity penalizes excessively liquid mega-cap stocks, which tend to offer lower alpha opportunity. At each 90-day rebalancing event the top $N = 100$ stocks by Q_t are selected and the MVO step (Max Sharpe) is solved over this reduced universe.

3.3.3 Results

The score-based approach produced a dramatic improvement in performance, as shown in Figure 3 and Table 3.

Table 3: Performance Summary for Score-Based Stock Selection Strategy.

Performance & Risk		Execution Statistics	
Total Return	1,436.69%	Trading Days	2,478
CAGR	32.03%	Rebalance Events	129
Ann. Volatility	15.83%	Total Trades	7,415
Sharpe Ratio	1.84	Avg Turnover	128.21%
Sortino Ratio	3.23	Total T-cost	44.32% (of initial capital)
Max Drawdown	-30.27%	Final Value	Rs. 76.8M

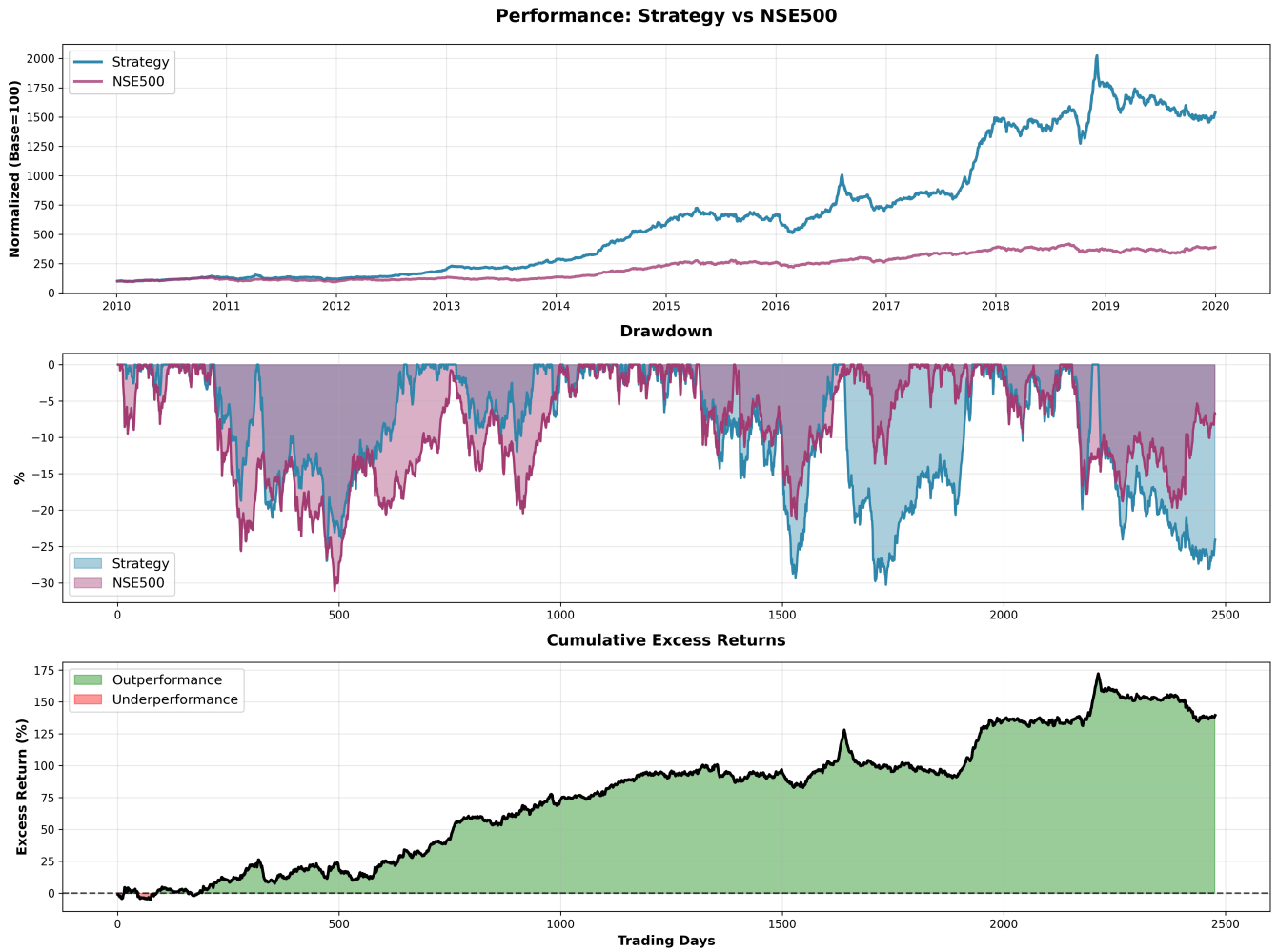


Figure 3: Score-Based Strategy vs NSE 500 Benchmark (2010–2020). *Top panel:* Normalized equity curves (base 100) showing the strategy (cyan) compounding to over 1500 while the NSE 500 (magenta) reaches approximately 400, an absolute alpha of roughly 1100 index points. *Middle panel:* Overlaid drawdown profiles; the strategy drawdown (blue shading) is generally shallower than the benchmark (magenta) during market-wide corrections. *Bottom panel:* Cumulative excess return (strategy minus benchmark) growing monotonically from zero to approximately 140%, confirming persistent and durable outperformance rather than a single lucky episode.

3.3.4 Discussion and Shortcomings

The score-based strategy represents a significant leap in performance: the Sharpe ratio improves from 1.42 (Approach 2) to **1.84**, and CAGR rises from approximately 19% to **32.03%**, corresponding to a total return of 1436.69%. The Sortino ratio of 3.23 is particularly notable, indicating that a large share of the volatility is *upside*.

However, three shortcomings motivated further refinement. First, the scoring mechanism treats all stocks as operating equally within the same market regime. A stock in a sector currently experiencing adverse headwinds may have a high individual momentum score, yet still underperform because the entire sector is de-rated. Second, the composite score used somewhat arbitrarily chosen weights, introducing a risk of parameter overfitting. Third, the average turnover per rebalance of 128.21% implies very high transaction costs (44.32% of initial equity over the period), suggesting the stock-selection mechanism is too aggressive in rotating the portfolio. This motivated the exploration of sector-level relative strength as a pre-filter.

3.4 Approach 4: Sectoral Relative Strength Analysis

3.4.1 Theoretical Foundation

Relative Strength (RS) in the sectoral context measures the performance of a sector index relative to a broad market benchmark. A sector with an RS ratio above 1.0 is outperforming the benchmark; one below 1.0 is underperforming. The formal definition for sector s on day t , normalized to the start of the observation window, is:

$$RS_{s,t} = \frac{P_{s,t}/P_{b,t}}{P_{s,0}/P_{b,0}}, \quad (14)$$

where $P_{s,t}$ is the price of sector index s and $P_{b,t}$ is the broad market benchmark price on day t .

The rationale for incorporating sector RS as a stock pre-filter is the **principle of sector rotation**: in any given market phase, certain stocks or sectors outperform structurally. Technology tends to outperform during risk-on environments and periods of innovation-driven growth, while Pharma tends to remain stable and resilient during market downturns.

3.4.2 Empirical Findings

Figure 4 presents the RS analysis for major NSE sectors against the Nifty 500 from 2010 to 2020. The strategy adopts a regime-aware stock selection approach that overweights outperforming stocks and underweights or excludes laggards, thereby filtering out stocks that underperform the NIFTY index. Specifically, it selects the top- K stocks based on RS as a filtering layer before applying the scoring step to select the final investable stocks, ensuring capital is allocated only to stocks that outperform the NIFTY Index.

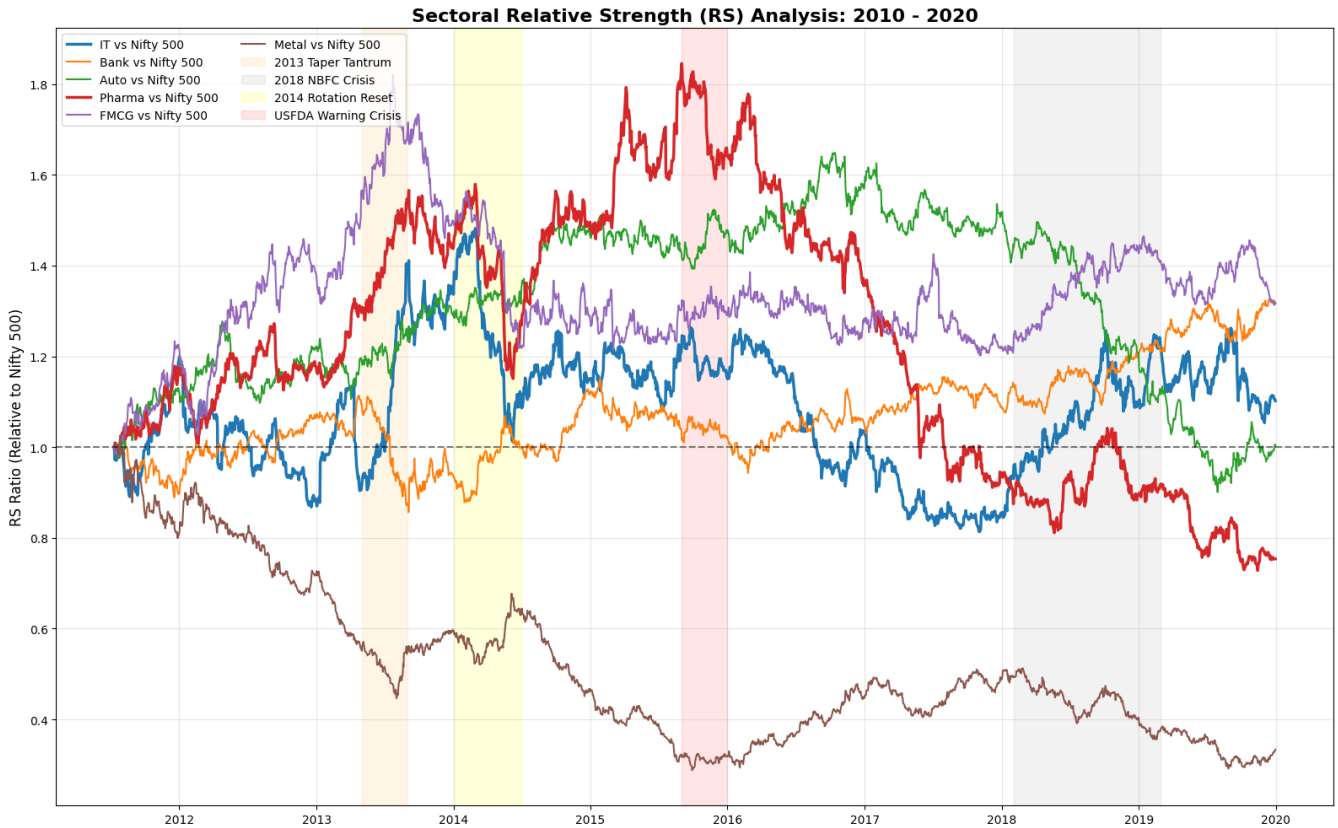


Figure 4: Sectoral Relative Strength Analysis (2010–2020). RS ratios normalized to 1.0 at the start of the 2011 period. The **Pharma sector (red)** exhibits the most dominant outperformance during the first half of the decade, peaking at an RS ratio near 1.8 in early 2015 before entering a structural decline due to regulatory headwinds. The **IT sector (blue)** and Pharma both experienced a sharp "relative crash" at the start of 2014 (shaded yellow region) as capital rotated aggressively into high-beta cyclicals like **Banking (orange)** and **Metals (brown)** ahead of the 2014 General Election. **Metals** remained the most consistent underperformer. **FMCG (purple)** maintained a high but plateauing relative strength, confirming its role as a steady defensive anchor.

3.5 Approach 5: Residual Momentum for Stock Selection

3.5.1 Formulation

To calculate a composite score for a stock that accounts for both trend strength and trend consistency, we make use of simple linear regression. The approach implemented here uses a log-linear regression of each stock's log-price against time to decompose performance into a trend component and idiosyncratic deviations:

Step 1: Log-Linear Regression. For stock i and a lookback window of T days:

$$\log P_{i,t} = \alpha_i + \beta_i \cdot t + \varepsilon_{i,t}, \quad t = 1, \dots, T, \quad (15)$$

where β_i is the log-linear trend slope (proportional to the continuously-compounded growth rate) and $\varepsilon_{i,t}$ are the residuals from the trend.

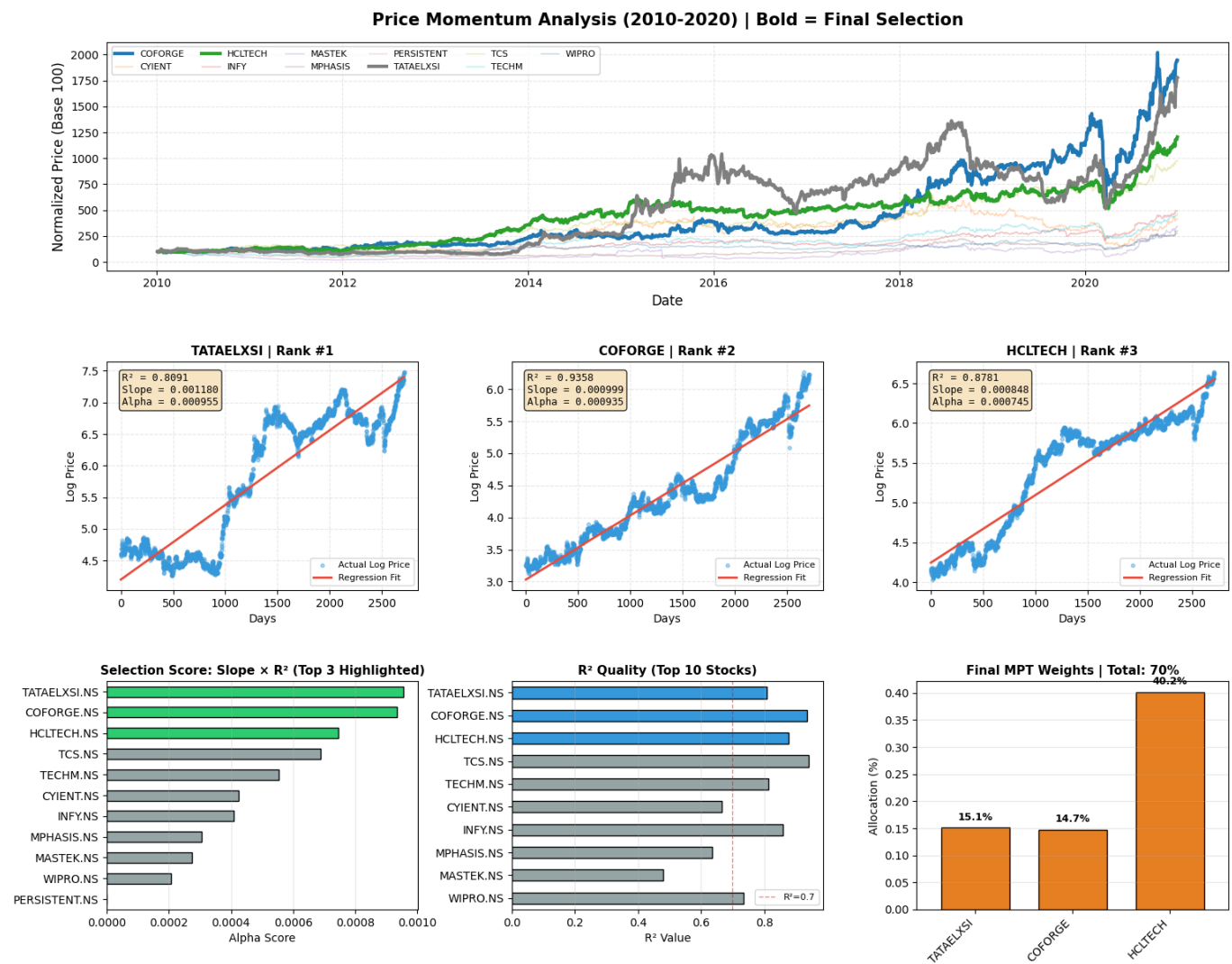


Figure 5: Residual Momentum Stock Selection — IT Sector Example. *Top panel:* Normalized price trajectories (base 100) for all stocks over the 10-year period; highlighted lines (bold) denote the three stocks ultimately selected: TATAELXSI, COFORGE, and HCLTECH. *Middle row:* Log-price versus time regression plots for the top-ranked stocks, with R^2 values of 0.81, 0.94, and 0.88 respectively; the high R^2 values confirm exceptionally consistent trajectories. *Bottom row (left):* Alpha scores (slope $\times$ R^2) for all stocks; the top-3 are clearly separated from the remaining names. *Bottom row (centre):* R^2 quality ranking illustrating that the minimum selection threshold of $R^2 > 0.7$ effectively eliminates noisy, choppy stocks from consideration. *Bottom row (right):* Final MPT allocation weights — HCLTECH receives the largest allocation at 40.2% due to its combination of strong trend (slope) and high R^2 .

Step 2: Trend Strength and R^2 . The coefficient of determination R^2 quantifies how consistently the stock follows its trend:

$$R_i^2 = 1 - \frac{\sum_t \hat{\varepsilon}_{i,t}^2}{\sum_t (\log P_{i,t} - \overline{\log P_i})^2}. \quad (16)$$

A high R^2 near 1.0 indicates that the stock's log-price closely tracks a smooth exponential growth trajectory — a stock that rises *steadily*, not erratically.

Step 3: Composite Alpha Score. The selection score combines slope and trend quality:

$$\alpha_i = \beta_i \cdot R_i^2. \quad (17)$$

This product rewards stocks that grow rapidly *and* consistently. A stock with a high β but low R^2 (high growth but an erratic path) scores lower than a stock with moderate β and very high R^2 (a steady compounder).

3.5.2 Illustrative Example and Results

Figure 5 presents the residual-based selection process applied to a sample of NSE IT sector stocks. The residual selection mechanism provides a cleaner signal than raw VAM scoring because it explicitly penalizes *noisy* momentum. The R^2 filter with threshold 0.7 ensures that only stocks demonstrating genuine, statistically reliable growth trajectories enter the final optimization. This has two important downstream benefits: it reduces the risk of selecting stocks whose recent price rise is a one-time jump rather than a durable trend, and it reduces the risk of parameter overfitting since this scoring mechanism is free of any hardcoded constants, unlike the VAM scoring.

4 Final Strategy

4.1 Strategy Architecture

The final strategy synthesises all insights from the five experimental approaches into a unified, fully automated pipeline. The architecture can be summarised as three sequential modules executed at each rebalancing event:

- 1. Universe Filtration via Relative Strength.** From the full universe of stocks, the top $K = 200$ stocks by rolling RS score are selected as the first filtering step. RS is computed as the 252-day return of each stock divided by the 252-day return of the broad Nifty 500 benchmark. This macro-level filter ensures the investable universe consists exclusively of stocks operating within outperforming sectors.
- 2. Residual Momentum Stock Selection.** Within the RS-filtered universe, the residual alpha score $\alpha_i = \beta_i \cdot R_i^2$ (Equation 17) is computed over a 252-day lookback. The top $N = 20$ stocks by α_i are selected as the final investable set.
- 3. Maximum Sharpe MVO with Rebalancing.** The Max Sharpe optimisation (Equation 6) is solved over the $N = 20$ selected stocks. Constraints of $w_{\min} = 0.02$ and $w_{\max} = 0.10$ are imposed to enforce diversification, avoiding excessive allocation of capital to a single outperforming stock. Portfolio rebalancing is triggered either by the **21-day static schedule** or by a **volume spike** (threshold $\theta = 1.35$), with a minimum inter-rebalance gap of seven days. A 30% fixed stop-loss on individual positions provides downside protection.

4.2 Hyperparameter Selection: Grid Search Analysis

The two critical hyperparameters — the number of RS stocks retained (K) and the final portfolio size (N) — were selected through an exhaustive grid search over $K \in \{100, 125, \dots, 300\}$ and $N \in \{5, 10, \dots, 40\}$. Three performance surfaces were evaluated: CAGR, Sharpe Ratio, and Maximum Drawdown.

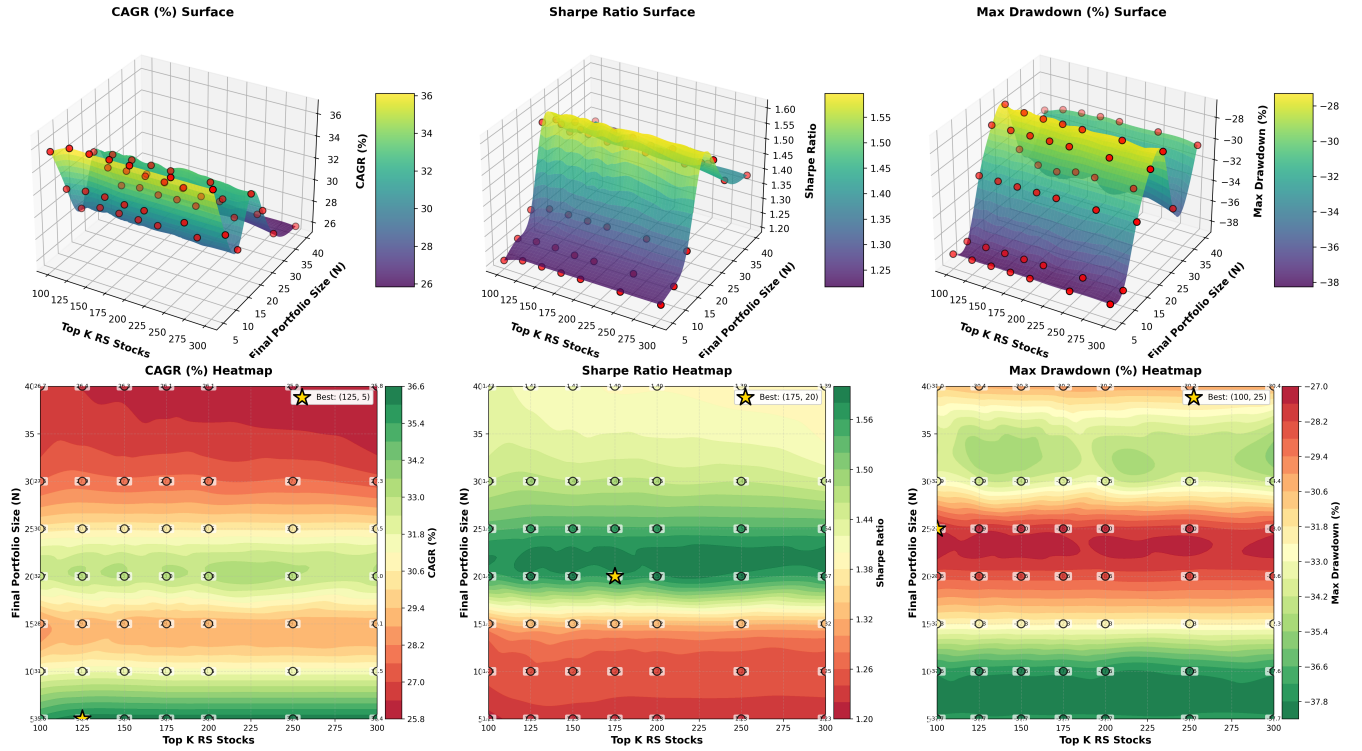


Figure 6: Grid Search Parameter Sensitivity. Top-down contour plots of the same three surfaces, enabling direct identification of the optimal operating point. *Left:* CAGR heatmap — the gold star marks the best CAGR configuration at $(K = 125, N = 5)$, but this corresponds to extreme concentration (only 5 stocks). *Centre:* Sharpe Ratio heatmap — the optimal risk-adjusted configuration is $(K = 175, N = 20)$, marked with a gold star; this configuration balances high momentum exposure with sufficient diversification. *Right:* Max Drawdown heatmap — the minimum drawdown configuration is $(K = 100, N = 25)$; lower K values consistently reduce drawdown, as the RS filter itself provides a quality screen that excludes high-volatility laggards. The chosen operating point of $(K = 200, N = 20)$ lies within the high-Sharpe region of the centre panel while maintaining a reasonable drawdown.

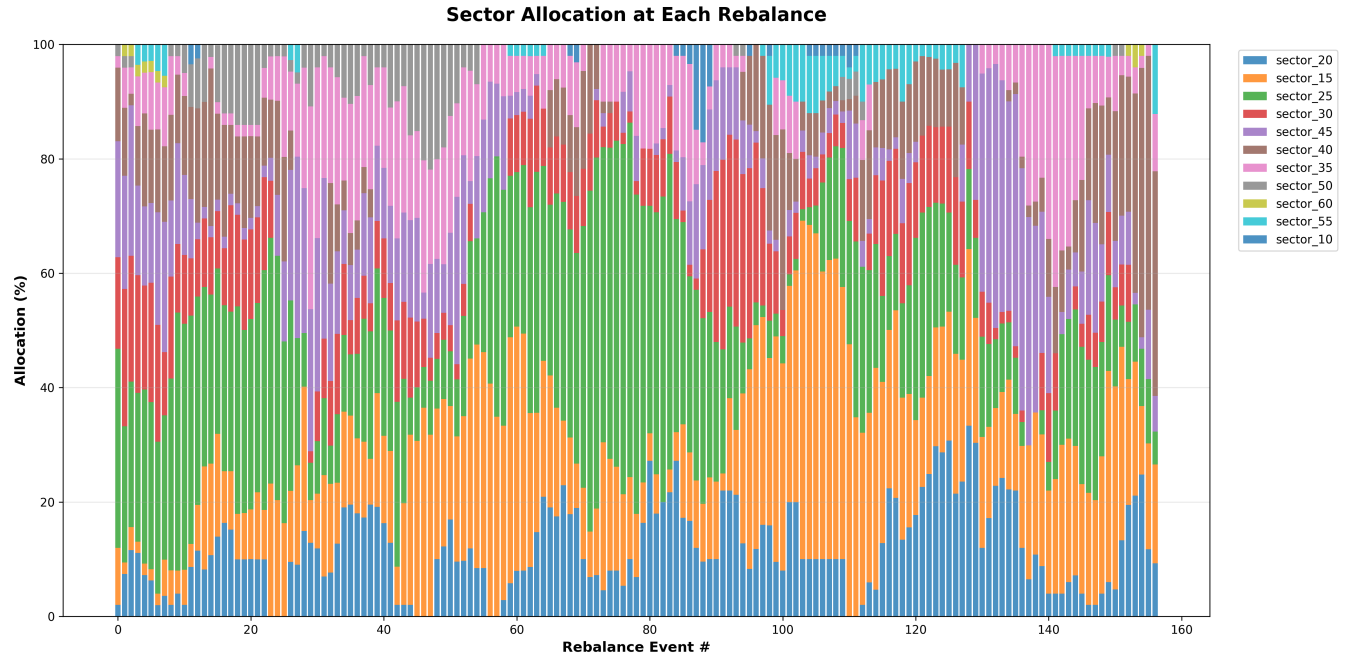


Figure 7: Sector Allocation at Each Rebalance Event (2010–2020). Each bar represents the portfolio sector composition at one of the 157 rebalancing events. The stacked colours correspond to GICS sector codes. The allocation is dynamic and frequently rotates between sectors, with no single sector permanently dominating the portfolio. Notably, periods of high sector-20 (Financials) and sector-25 (Consumer Discretionary) concentration align with bull market phases, while more defensive allocations emerge during stress periods. The breadth of sector participation across all 157 rebalances confirms that the RS pre-filter successfully cycles through leadership rotations rather than anchoring to a fixed sector bet.

The grid search reveals several important structural insights. CAGR and Sharpe are more sensitive to N (with a sweet spot around 15–25 stocks). Drawdown is also driven by N , with very small portfolios of fewer than 10 stocks exhibiting substantially worse worst-case losses. All three metrics are almost independent of K . The final choice of ($K = 200, N = 20$) reflects a deliberate balance: it lies within one standard deviation of the Sharpe-optimal region while avoiding the extreme concentration risk of the very best-CAGR configurations.

4.3 Sector Diversification Across Rebalances

A key risk in momentum-driven strategies is implicit sector concentration — the strategy may effectively become a bet on a single high-momentum sector, amplifying drawdown when that sector corrects. Figure 7 demonstrates that the final strategy maintains meaningful sector diversification throughout the backtesting period. This diversification of the portfolio is attributable to two main reasons:

- **RS-based stock filtering:** Selects only stocks from the top-performing sectors at any given time.
- **Weight constraints:** During optimization, a constraint on the weights prevents any single stock from holding more than 10% of the portfolio.

4.4 Final Strategy Performance

Figure 8 presents the complete performance panel for the final strategy, and Figure 9 provides the benchmark comparison. Table 4 summarizes the comprehensive metrics. Table 5 summarizes the parameters used during the backtesting of the final strategy.

Table 4: Comprehensive Performance and Benchmark Comparison — Final Strategy (2010–2020).

Core Performance Metrics			
CAGR	37.51%	Total Return	1,569.19%
Annualised Volatility	21.47%	Sharpe Ratio	1.59
Sortino Ratio	2.55	Max Drawdown	−26.96%
Win Rate (daily)	57.73%	Trading Days	2,227
Execution Statistics			
Rebalance Events	157	Total Trades	3,606
Stop Loss Triggers	61	Avg Turnover	44.47%
Total T-cost	146.06% (of initial capital)	Initial Capital	Rs. 5,000,000
Final Portfolio Value	Rs. 83,459,735		
Strategy vs Nifty 500 Benchmark			
Strategy CAGR	37.51%	Nifty 500 CAGR	8.28%
Strategy Sharpe	1.59	Nifty 500 Sharpe	0.60
Information Ratio	1.511	Alpha	1,468.33%
Up Capture	112.54%	Down Capture	85.85%
Strategy Max DD	−26.96%	Nifty 500 Max DD	−28.65%
Rolling Outperformance vs Nifty 500			
1-Year Avg	+35.53%	1-Year Worst	−17.43%
3-Year Avg	+201.14%	3-Year Worst	+38.21%
5-Year Avg	+583.16%	5-Year Worst	+176.69%

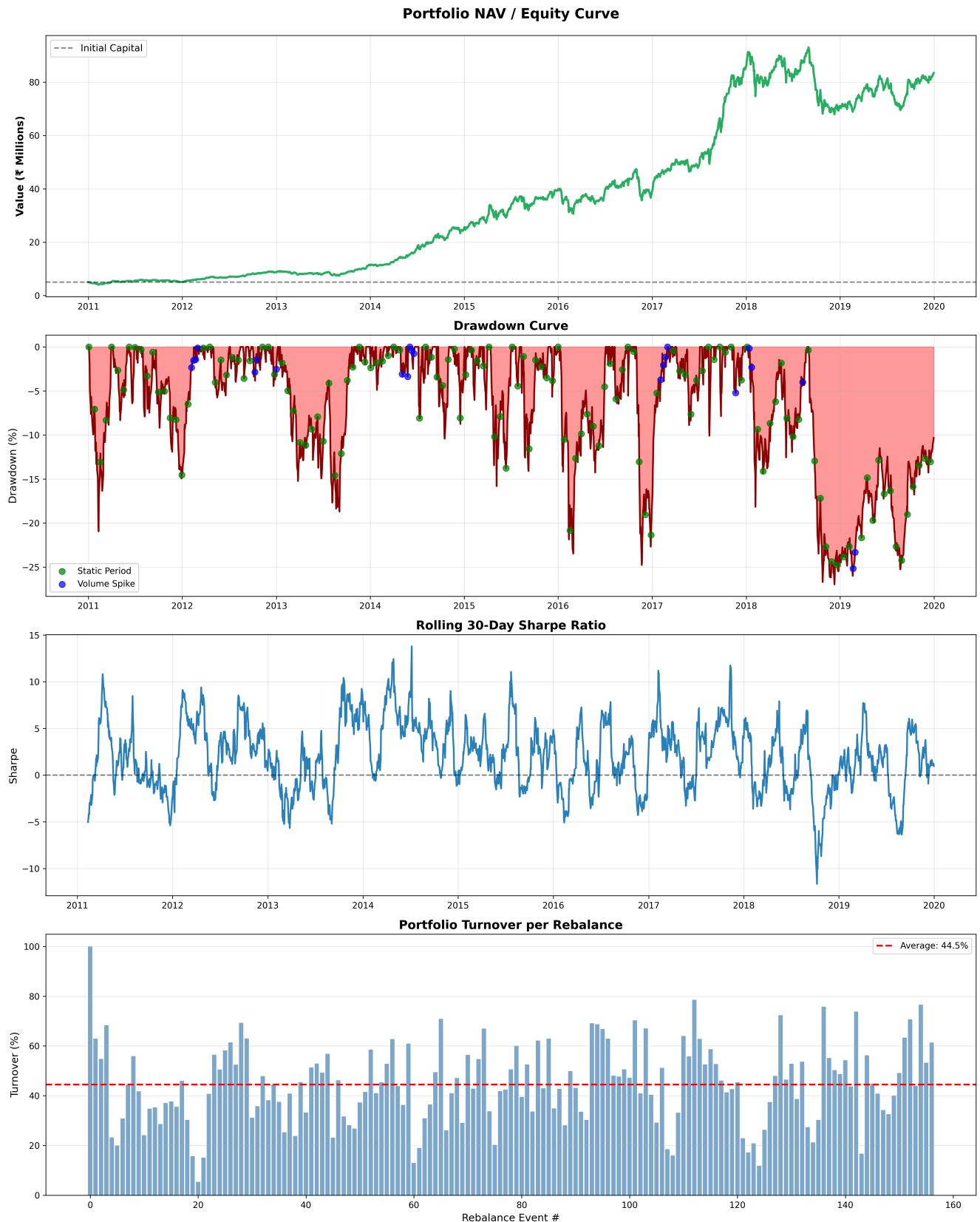


Figure 8: Final Strategy Backtest Performance Panel (2010–2020). *Top:* Portfolio NAV (equity curve) growing from Rs. 5M initial capital to Rs. 83.5M terminal value. The curve exhibits a characteristic exponential growth pattern, with compounding effects becoming visibly dominant from 2014 onwards. *Second panel:* Drawdown curve with rebalancing events marked: green dots indicate 21-day static triggers and blue dots indicate volume-spike triggers. The maximum drawdown of -26.96% is clearly visible in the 2018–2019 period, concentrated around the global risk-off episode. Volume-triggered rebalances (blue) cluster appropriately around the deepest drawdown points, confirming the mechanism is functioning as intended. *Third panel:* Rolling 30-day Sharpe Ratio, illustrating the strategy’s time-varying risk-adjusted performance. Despite the significant 2018 drawdown, the Sharpe rarely remains negative for extended periods, indicating rapid recovery. *Bottom:* Rebalance-by-rebalance turnover, averaging 44.47% (red dashed line), well below the 128.21% of Approach 3.

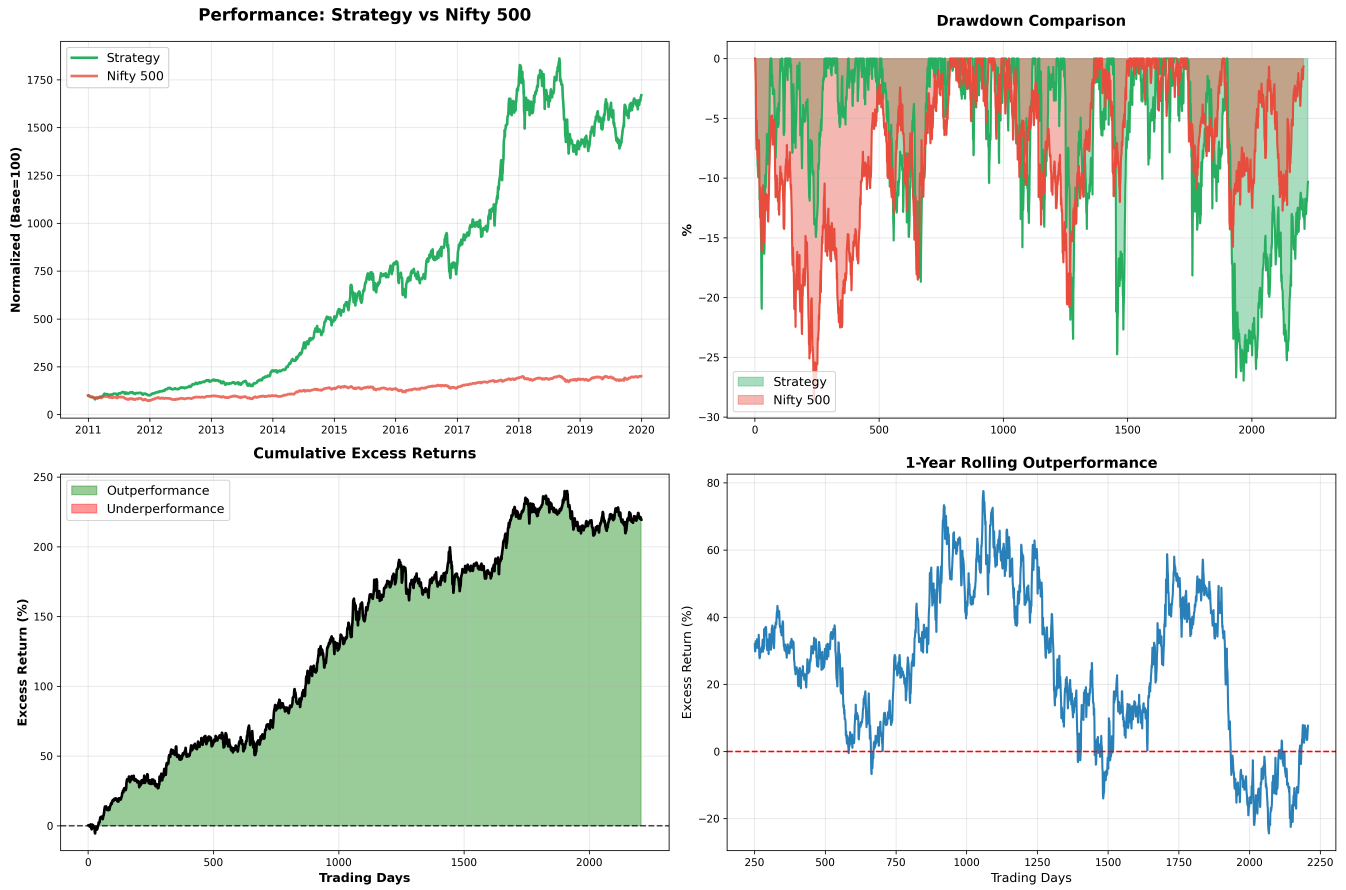


Figure 9: Final Strategy vs Nifty 500 Benchmark (2010–2020). *Top left:* Normalised performance (base 100) illustrating the strategy (green) reaching approximately 1669 while the Nifty 500 (red/orange) ends at approximately 201, a relative outperformance of more than $15\times$. *Top right:* Drawdown comparison — the strategy’s drawdown profile (green shading) is generally tighter than the benchmark (red shading) outside of the 2018–2019 stress period. *Bottom left:* Cumulative excess return growing smoothly to over 200%. *Bottom right:* 1-year rolling outperformance, averaging 35.53% per year with the worst one-year underperformance being -17.43% , representing a brief reversal in 2019–2020.

Table 5: Final Strategy Configuration Parameters.

Parameter	Value	Description
Scoring lookback	252 days	Rolling window for RS and alpha score computation
Optimisation lookback	252 days	Covariance matrix estimation window
Top- K RS stocks	200	Universe size after RS filter
Final portfolio size N	20	Number of stocks after residual selection
Min weight $w_{\min}$	2%	Lower bound on individual stock weight
Max weight $w_{\max}$	10%	Upper bound on individual stock weight
Static rebalance period	21 days	Base rebalancing frequency
Volume spike threshold θ	1.35	Trigger multiplier for volume-based rebalancing
Min inter-rebalance gap	7 days	Minimum days between any two rebalances
Transaction cost	0.268%	Per-side round-trip transaction cost
Stop-loss level	30%	Fixed stop-loss on individual positions
Initial capital	Rs. 5,000,000	Starting portfolio value

5 Discussion of Results

5.1 Performance Highlights

The final strategy delivers **exceptional absolute and risk-adjusted performance** relative to both the benchmark and the earlier experimental approaches. The CAGR of 37.51% versus the benchmark CAGR of 8.28% represents a **29.23% annual**

alpha over a decade — a magnitude that, if maintained, would compound Rs. 1 into approximately $15\times$ more wealth than the benchmark over the same period.

The **Sharpe Ratio of 1.59** against the benchmark's 0.60 confirms that the outperformance is not merely a consequence of taking more risk; the strategy generates nearly three times the risk-adjusted excess return per unit of volatility. This is further validated by the **Information Ratio of 1.511**.

The **capture ratio analysis** is particularly revealing: the strategy captures 112.54% of the benchmark's up-moves while capturing only 85.85% of its down-moves. This asymmetry — capturing more of rallies and less of declines — is the hallmark of a quality momentum strategy and partly explains why the strategy's maximum drawdown (-26.96%) is actually lower than the benchmark's (-28.65%) despite a CAGR nearly five times larger.

5.2 Alpha Persistence

The rolling performance analysis provides strong evidence of alpha persistence rather than a one-time lucky period. Over any **3-year rolling window**, the strategy outperformed the benchmark by an average of 201%, with the *worst 3-year period still delivering 38.21% outperformance* — indicating the strategy has never underperformed over any 3-year horizon within the backtesting period. Over 5-year windows the average outperformance rises to 583% with a worst case of 176.69%, confirming the compounding benefits of sustained alpha.

5.3 Turnover and Transaction Costs

One of the most significant practical improvements from Approach 3 to the final strategy is the reduction in portfolio turnover from 128.21% to 44.47% per rebalance. This directly reflects the value of the RS pre-filter: by retaining only stocks already in an upward momentum regime, the strategy avoids the costly rotation of poor-performing names that characterized the VAM-only approach. Despite this, the cumulative transaction cost over the period amounts to 146.06% of initial capital — a substantial figure that highlights the importance of cost modelling in realistic backtesting. Net of costs, the strategy remains highly attractive, but live implementation would benefit from careful execution optimization.

5.4 Drawdown and Risk Management

The 30% stop-loss mechanism triggered 61 times over the backtesting period, providing meaningful downside protection on individual positions. The deepest drawdown period (2018–2019) coincides with the global risk-off episode driven by US–China trade tensions and rising interest rates, confirming that the strategy is not immune to macro-driven market-wide sell-offs. However, the volume-triggered rebalancing mechanism clustered additional repositioning events around this drawdown, limiting the depth to -26.96% .

6 Conclusion

This report has documented the systematic development of a quantitative long-only equity strategy on the NSE 500 universe. Starting from a static Mean-Variance Optimization baseline, five successive enhancements were introduced: dynamic rebalancing (static and volume-triggered), composite stock scoring using volatility-adjusted momentum, trend, and liquidity features, sectoral Relative Strength pre-filtering, and residual momentum selection via log-linear regression. The final strategy integrates all these components and achieves a CAGR of 37.51%, a Sharpe Ratio of 1.59, and a total return of 1,569.19% over the decade, growing an initial capital of Rs. 5M to Rs. 83.5M.

Each iterative step yielded measurable improvements in both absolute return and risk-adjusted metrics, validating the research methodology. The use of RS as a macro pre-filter simultaneously improved returns and reduced portfolio turnover by approximately 65%, demonstrating that better stock selection reduces not only signal noise but also implementation cost.